Spectrum transition through the sonic point: are the tools ready?

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Abstract

The goal is to follow the spectrum of the magnetic-nozzle eigenvalue problem (thesis Eq. 3.6) as the throat Mach number v_m is moved through 1. This note checks the two available tools against that goal. (i) InfBeyn as used for the non-transonic flows works in principle for every $v_m \neq 1$, but on the plain Chebyshev grid its cost explodes as $v_m \to 1$: the resolvent solutions develop a layer of width $\sim |1 - v_m|$ at the throat, the degree per solve grows like $100/|1 - v_m|$, and the scan at $v_m = 0.99$ does not resolve with $N = 4096$. A one-line change, a sinh map clustering the grid at the throat, removes the problem: every case down to $|1 - v_m| = 10^{-4}$ resolves with $N \leq 1024$, and full certified scans take seconds to a minute or two. (ii) The Frobenius idea (select the regular solution at the sonic throat) is correct and is what makes the transonic problem well posed, but the thesis implementation is not accurate enough: its series coefficient c_2 drops terms, the profile derivatives come from finite differences, and the shooting search is not exhaustive. Re-solving the accelerating transonic flow with InfBeyn and the regularity condition imposed as a collocation row confirms the first thesis eigenvalue to 10^{-4}, corrects the second by 0.07, and shows that thesis modes 3–8 are not eigenvalues. (iii) The eigenvalues move at a speed $|d\omega/dv_m|$ that grows as $v_m \to 1$ while their spacing stays $O(1)$; a v_m grid uniform in $\ln|1 - v_m|$ with step $\Delta v_m \approx 0.1\,|1 - v_m|$ (about 23 points per decade on each side) tracks every mode in the window.

1 Problem and what "crossing the sonic point" means

The eigenvalue problem is (thesis Eq. 3.6)

$$\begin{aligned} T(\omega)\tilde{v} \equiv \omega^2\tilde{v} + 2i\omega\left(v_0\tilde{v}' + v_0'\tilde{v}\right) + (1 - v_0^2)\tilde{v}'' \\ - \Big(3v_0 + \frac{1}{v_0}\Big)v_0'\tilde{v}' - \Big[\Big(1 - \frac{1}{v_0^2}\Big)v_0'^2 + \Big(v_0 + \frac{1}{v_0}\Big)v_0''\Big]\tilde{v} = 0 \end{aligned} \tag{1}$$

on $z \in [-1, 1]$ with fixed-open ($\tilde{v}(-1) = \tilde{v}'(1) = 0$) or Dirichlet boundaries, and

$$v_0(z) = \big[- W_k\big(- (B/B_m)^2 v_m^2 e^{-v_m^2}\big)\big]^{1/2},$$

with the same field as the non-transonic study ($B_0 = 1$, $R = 1.5$, $\Delta = 1/3$).

Figure 1 shows the family. For $v_m < 1$ the whole profile is subsonic ($k = 0$) and for $v_m > 1$ supersonic ($k = -1$). The two families meet at $v_m = 1$ in a profile with a corner at the throat: $1 - v_0 \approx 2.32|z|$ for $v_m \to 1^-$, and $v_0 - 1 \approx 2.32|z|$ for $v_m \to 1^+$. The smooth accelerating transonic profile is a third branch through the same X-point; it is not the limit of either family. Moving v_m through 1 therefore means passing through a singular limit, and on each side the coefficient $1 - v_0^2$ of $\tilde{v}''$ is of size $|1 - v_m^2|$ in a throat layer of width $\sim |1 - v_m|$ (Fig. 1, right).

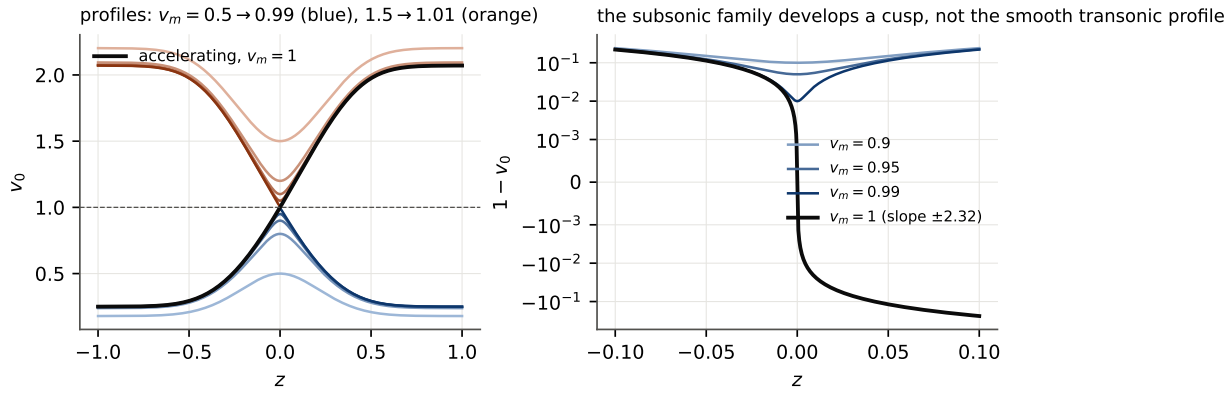


Figure 1: Left: profiles for $v_m = 0.5, 0.8, 0.9, 0.95, 0.99$ (blue) and $1.5, 1.2, 1.1, 1.05, 1.01$ (orange), with the smooth accelerating transonic profile (black). Right: $1 - v_0$ near the throat. The subsonic family approaches a cusp, while the transonic profile crosses $v_0 = 1$ with slope 2.32.

2 Is InfBeyn ready for $v_m \to 1$?

On the plain grid it is not. Every resolvent solve $T(\omega)^{-1}V$ in a scan must resolve a layer at the throat. There the equation is nearly singular, and solutions oscillate on a scale $\sim (1 - v_0)/|\omega|$. The adaptive Chebyshev solver doubles N until the tail is below 10^{-12}. Figure 2 (orange) shows the degree it reaches: $N = 256$ at $v_m = 0.5$, 1024 at 0.9, 2048 at 0.95, and unresolved at 4096 for 0.99, close to $N \approx 100/|1 - v_m|$. The cost per dense solve grows like N^3 and the memory like N^2 per worker. Table 1 shows the effect on a full scan of the fixed-open window: 9 s at $v_m = 0.5$ and 376 s at $v_m = 0.9$, with five estimates left unpolished. The $v_m = 0.95$ scan ran for more than 40 minutes and exhausted the 15 GB of the workstation (12 workers, each factoring 4097×4097 complex matrices) before it was stopped. At $v_m = 0.99$ no solve resolves at all. Extrapolating, a plain-grid sweep to $|1 - v_m| = 10^{-2}$ is a cluster job, and 10^{-3} is out of reach.

With a throat-clustered grid it is. Replacing $x \in [-1, 1]$ by $z = \sinh(Lx)/\sinh L$, which compresses the grid at the throat by $L/\sinh L$, with

$$L = 1 + 1.5 \ln \frac{1}{|1 - v_m|} \qquad (|1 - v_m| < 0.3, \text{ else } L = 0), \tag{2}$$

keeps the degree of a single resolvent solve between 128 and 1024 for all $|1 - v_m| \geq 10^{-4}$ (Fig. 2, blue). Only `T` and `bc` change: the derivative matrices become $D_z = D_x/g'$ and $D_{zz} = D_x^2/g'^2 - (g''/g'^3)D_x$ (`mapped_problem` in `common.py`); `inf_beyn.py` is untouched. The bottom half of Table 1 gives the full scans: every eigenvalue in the window is certified (halo $> 10\,\gamma_{256}$ for every eigenvalue, recorded in `data/sweep.log`) for $v_m = 0.5$–0.99 and 1.005–1.5, in 4 to 90 s per scan. The runs at $v_m = 0.995$, 0.999 and 1.001 were not completed (see the memory caveat in Sec. 4). Closer to 1 two costs grow: the number of eigenvalues in the window, so the scan subdivides more and needs more solves (3680 at 0.5, 16430 at 0.99); and Newton polishing of a few eigenfunctions, which still climbs to $N = 4096$ ("max N" column).

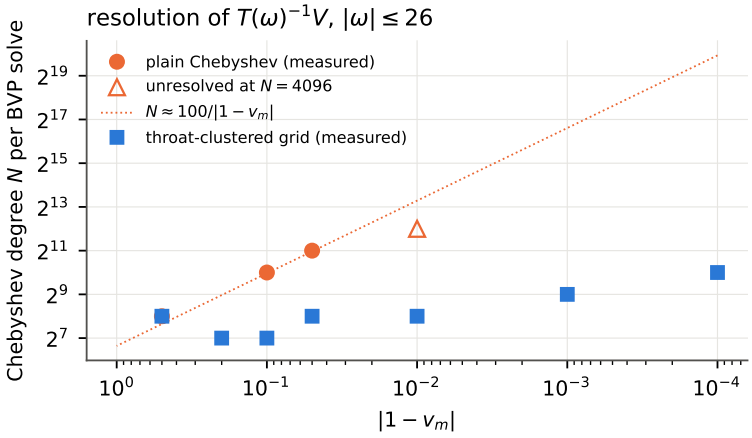


Figure 2: Chebyshev degree reached by the adaptive BVP solver for $|\omega| \le 26$, plain grid (orange) and throat-clustered grid (blue).

v_m	grid	scan [s]	BVP solves	max N	#eig	unpolished	unresolved tiles
0.5	plain (thesis run)	9	3680	512	18	0	0
0.9	plain	376	4559	4096	25	5	0
0.5	mapped, $L = 0.0$	9	3680	512	18	0	0
0.8	mapped, $L = 3.4$	4	2863	256	22	0	0
0.9	mapped, $L = 4.5$	7	5941	512	25	0	0
0.95	mapped, $L = 5.5$	23	11281	512	28	0	0
0.98	mapped, $L = 6.9$	37	12850	512	31	0	0
0.99	mapped, $L = 7.9$	87	16430	4096	33	1	0
1.5	mapped, $L = 0.0$	32	413	4096	7	1	0
1.3	mapped, $L = 0.0$	47	1921	4096	9	2	0
1.1	mapped, $L = 4.5$	4	2926	512	13	0	0
1.05	mapped, $L = 5.5$	8	2863	512	15	0	0
1.02	mapped, $L = 6.9$	13	3679	512	18	0	0
1.01	mapped, $L = 7.9$	54	5501	4096	21	1	0
1.005	mapped, $L = 8.9$	62	7637	4096	24	2	0

Table 1: Full InfBeyn scans of the fixed-open window $\mathrm{Re}\,\omega \in [-0.5, 26]$, $\mathrm{Im}\,\omega \in [-3, 3]$ on 12 cores (4 for the second batch of mapped runs). "max N" is the degree of the longest eigenfunction. The plain-grid scan at $v_m = 0.95$ is missing: it ran out of memory (see text).

3 Is the Frobenius method good for the transonic flow?

The idea, yes. At $v_0 = 1$ the throat is a regular singular point of Eq. (1). Its indicial exponents are $r = 0$ (the regular solution) and

$$r_s = -1 + \frac{i\omega}{v_0'(0)}, \qquad \mathrm{Re}\, r_s = -1 - \frac{\mathrm{Im}\,\omega}{v_0'(0)}, \quad v_0'(0) = 2.3238. \tag{3}$$

Requiring regularity is what turns the transonic problem into a well-posed eigenvalue problem, and it is the correct physical condition. For $\mathrm{Im}\,\omega < -v_0'(0)$ the second solution is also bounded, so "regular" must then be read as "analytic at the throat". This is a modeling choice that the thesis makes implicitly.

The thesis implementation, no. Three issues limit its accuracy:

1. The recurrence linearizes v_0 and drops the $O(z)$ parts of the first- and zeroth-order coefficients. That leaves c_0 and c_1 exact but makes c_2 wrong by $O(1)$. For mode 1, the thesis value is $c_2 = -0.82 + 0.51i$ and the exact one is $-1.32 + 0.05i$.

2. The profile and its derivatives come from finite differences on 501 points and linear interpolation. The integration switches to the frozen series derivatives for $|z| < 0.01$.

3. The search is a Newton iteration from 11 starting points on the real axis. It is not exhaustive and can converge to non-eigenvalues of the discretized shooting function.

Redoing the shooting carefully, with exact series coefficients started at $z = -10^{-3}$, an exact profile and DOP853 at 10^{-10}, moves the thesis eigenvalues by 9×10^{-5} (mode 1), 0.07 (mode 2) and 1.5–4 (modes 3–8). The last group no longer converges to the same points.

A better tool: InfBeyn with the regularity condition as a collocation row. On the half nozzle $[-1, 0]$ (mapped to $[-1, 1]$), the Chebyshev point at the throat is where $1 - v_0^2 = 0$. Imposing the ODE there, without a boundary condition, is exactly the Frobenius condition $c_1 = -R_0 c_0 / Q_0$. The only boundary row is $\tilde{v}(-1) = 0$. Collocation picks the analytic solution automatically, and every solve resolves with $N = 64$. This is `TransonicInfBeyn` in `common.py`: it overrides the BVP matrix, the right-hand side handling and the Newton derivative so that the throat row is treated as an equation, not a boundary row. The scan of $\mathrm{Re}\,\omega \in [-0.5, 22]$, $\mathrm{Im}\,\omega \in [-6, 3]$ takes about 110 s (Fig. 3, Table 2). It finds exactly two eigenvalues above $\mathrm{Im}\,\omega \approx -5.4$:

$$\omega_1 = 3.687406 - 1.003933i, \qquad \omega_2 = 6.943383 - 3.181913i,$$

both certified ($\gamma_{256} \sim 10^{-9}$, halos 0.8 and 0.03), and both equal to the careful shooting values to 10^{-5}. The region where thesis modes 3–8 sit ($\mathrm{Im}\,\omega \approx -3.8$) is resolved and contains no eigenvalue. Below $\mathrm{Im}\,\omega \approx -5.4$ the scan does not resolve, and γ_n falls by an order of magnitude for every unit of $\mathrm{Im}\,\omega$, so the pseudospectrum fills that strip. The four careful-shooting roots at $\mathrm{Im}\,\omega \approx -5.4$ to -5.5 lie in it and cannot be certified.

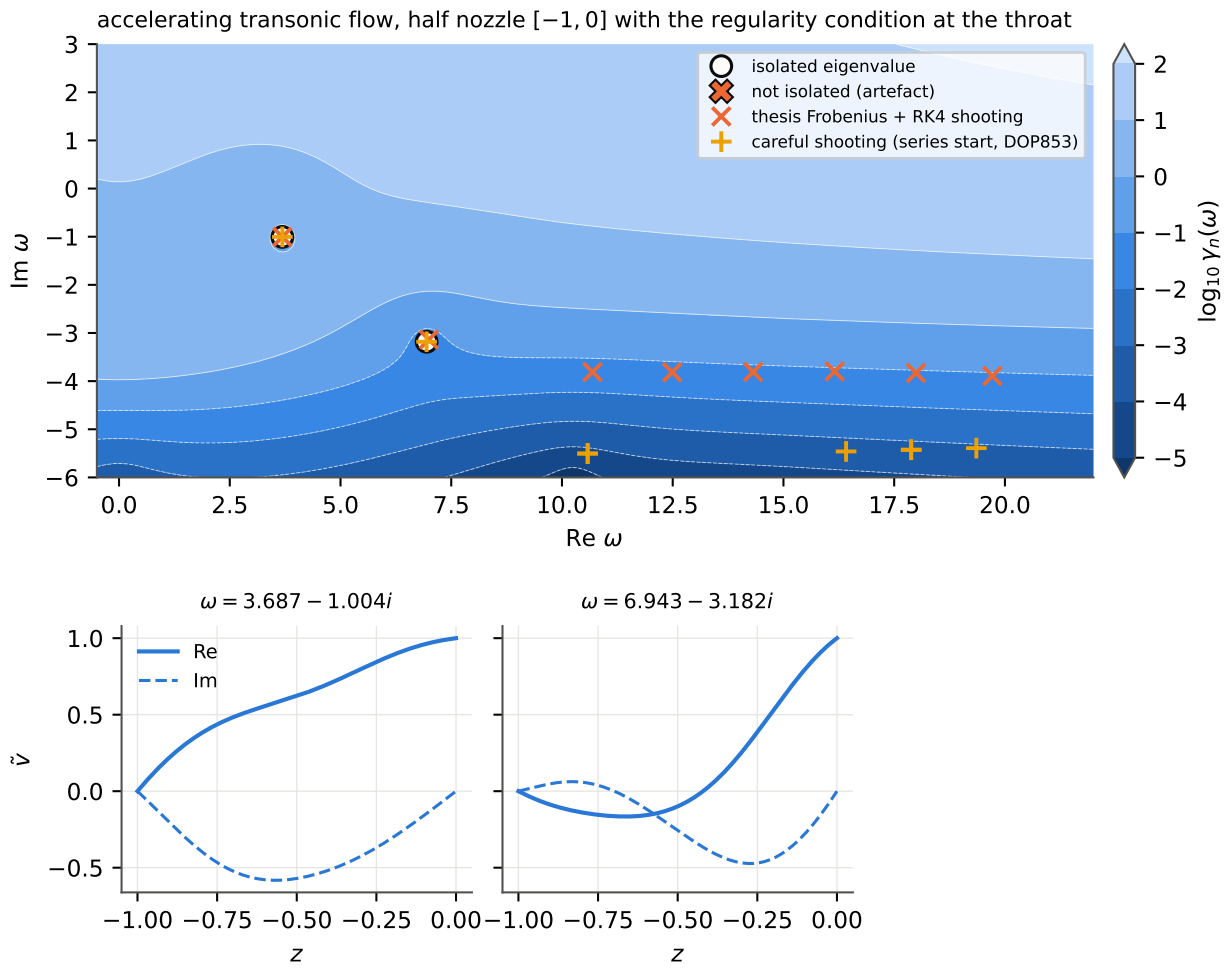


Figure 3: Accelerating transonic flow. Top: InfBeyn eigenvalues on the pseudospectrum, with the thesis shooting eigenvalues (orange crosses) and careful re-shooting from them (yellow plus signs). Bottom: the two eigenfunctions on the subsonic half; they are extended through the throat by the same regular solution.

n	ω_n (InfBeyn)	γ_{256}	halo	thesis shooting	$\lvert\Delta\rvert$
1	$3.687406 - 1.003933i$	2.2×10^{-9}	8.2×10^{-1}	$3.6874 - 1.0040i$	8.8×10^{-5}
2	$6.943383 - 3.181913i$	2.6×10^{-9}	3.2×10^{-2}	$6.9954 - 3.1360i$	6.9×10^{-2}

Table 2: Accelerating transonic flow: certified eigenvalues and the nearest thesis value.

Verdict. Use the Frobenius condition, but impose it inside InfBeyn rather than by shooting. The transonic $v_m = 1$ problem is then a first-class member of the same toolchain, at a cost comparable to a non-transonic scan. One consequence matters for the transition study: the accelerating transonic flow is damped in the window, with largest growth rate $\mathrm{Im}\,\omega_1 = -1.00$. The subsonic family stays weakly unstable as $v_m \to 1^-$ and the supersonic family strongly unstable as $v_m \to 1^+$ (Sec. 4). The transonic spectrum is not the limit of either.

4 How dense must the v_m grid be?

Measurement. Chebyshev collocation on the throat-clustered grid ($N = 300$, keeping only eigenvalues reproduced at $N = 400$ to 10^{-6}) is cheap enough to follow the whole window over 94 values of v_m in $[0.5, 1.5]$. It took 21 minutes for fixed-open and 22 minutes for Dirichlet. It is used here only to measure how the spectrum moves; the certified eigenvalues come from InfBeyn. Figure 4 shows the spectra and, on the right, the median speed $|d\omega/dv_m|$ of the eigenvalues together with their minimum spacing.

- The spacing stays $O(1)$: $1.4 \to 0.5$ on the subsonic side, $3 \to 0.7$ on the supersonic side.
- The speed grows as $v_m \to 1$: the median goes from 3 at $v_m = 0.5$ to ~ 200 at $|1 - v_m| = 10^{-3}$, and the fastest mode is two to five times faster.
- Requiring that no eigenvalue moves more than a quarter of the spacing per step gives $\Delta v_m \approx 0.11\,|1 - v_m|$ on both sides. Examples: 0.011 at $v_m = 0.9$, 0.001 at 0.99, 0.0012 at 1.01, 0.013 at 1.1.

Recommendation. Use a grid uniform in $\ln|1 - v_m|$ with ratio ≈ 0.9 between consecutive values of $|1 - v_m|$, which is about 23 points per decade on each side:

range of $\lvert 1 - v_m \rvert$	points per side	both sides
$0.5 \to 10^{-2}$	40	80
$0.5 \to 10^{-3}$	62	124
$0.5 \to 10^{-4}$	85	170

A grid uniform in v_m would need $\Delta v_m \approx 10^{-4}$ throughout, about 10^4 points, to reach $|1 - v_m| = 10^{-3}$. Tracking can be done by nearest neighbour between consecutive InfBeyn scans, or by polishing the previous eigenvalues with `InfBeyn.polish`, with a full scan every few steps to catch modes entering through the window edges.

Cost. From Table 1, a certified scan on the mapped grid costs 4–25 s for $|1 - v_m| \geq 0.05$ and 50–90 s at $|1 - v_m| = 0.005$–0.01. The 80-point grid to $|1 - v_m| = 10^{-2}$ is about 30 minutes on this 12-core workstation. Below 10^{-2} the current implementation is limited by memory, not by time:

- `scan()` keeps every resolvent solve $T(z)^{-1}V$ (24 columns of Chebyshev coefficients) in a dictionary until the scan ends. At $v_m = 0.995$ the parent process reached 10 GB after 10 minutes and had to be stopped. Only the edge integrals and the node norms are needed once an edge is processed, so discarding the solutions after `_segment_integrals` would remove this.
- When `polish` does not converge it doubles N up to `n_max`$= 4096$ in every worker. With 12 workers this also exhausted the 15 GB. Use 4–6 workers or a lower `n_max` near $v_m = 1$.

With the cache fix, a rough extrapolation of Table 1 (solves growing like $|1 - v_m|^{-0.4}$) puts the 124-point grid to 10^{-3} at one to two hours. On the plain grid it is not feasible (Sec. 2).

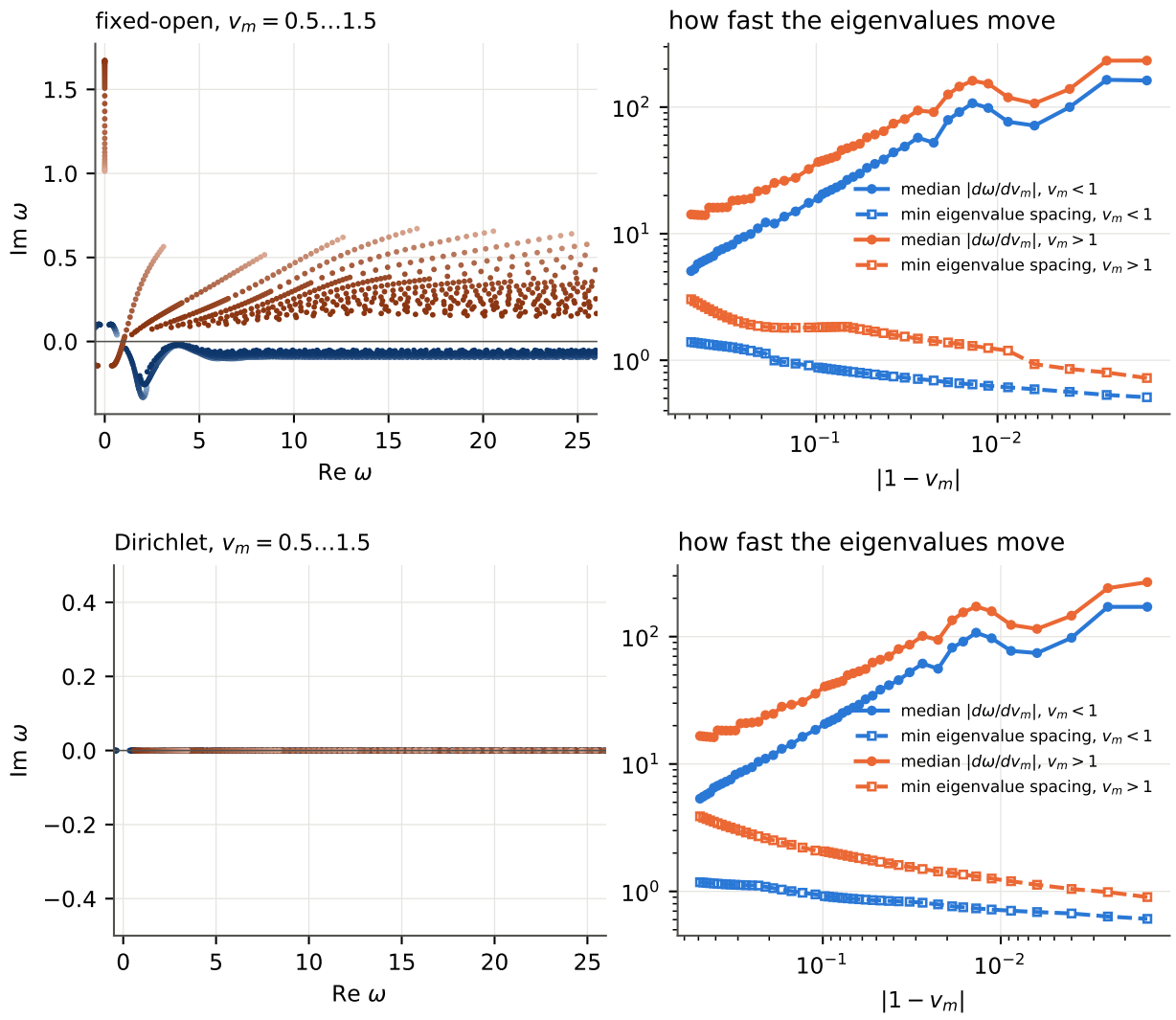


Figure 4: Collocation continuation in v_m (fixed-open top, Dirichlet bottom). Left: all converged eigenvalues for $v_m = 0.5 \ldots 0.999$ (blue, darker closer to 1) and $1.001 \ldots 1.5$ (orange). Right: median eigenvalue speed and minimum spacing against $|1 - v_m|$.

First look at the transition (fixed-open). The largest growth rate on the subsonic side rises slowly, from 0.039 at $v_m = 0.5$ to about 0.10 at $v_m = 0.99$ (the unstable ground mode). On the supersonic side it rises from 1.01 at $v_m = 1.5$ to 1.67 just above $v_m = 1$ (the purely imaginary mode). The jump in growth rate across $v_m = 1$ is therefore a factor of about 16. The subsonic spectrum gets denser as $v_m \to 1^-$, with 18 eigenvalues in the window at 0.5 and 39 at 0.998. With Dirichlet boundaries all eigenvalues stay real on both sides. Figures 5 and 6 show the certified InfBeyn spectra from the sweep.

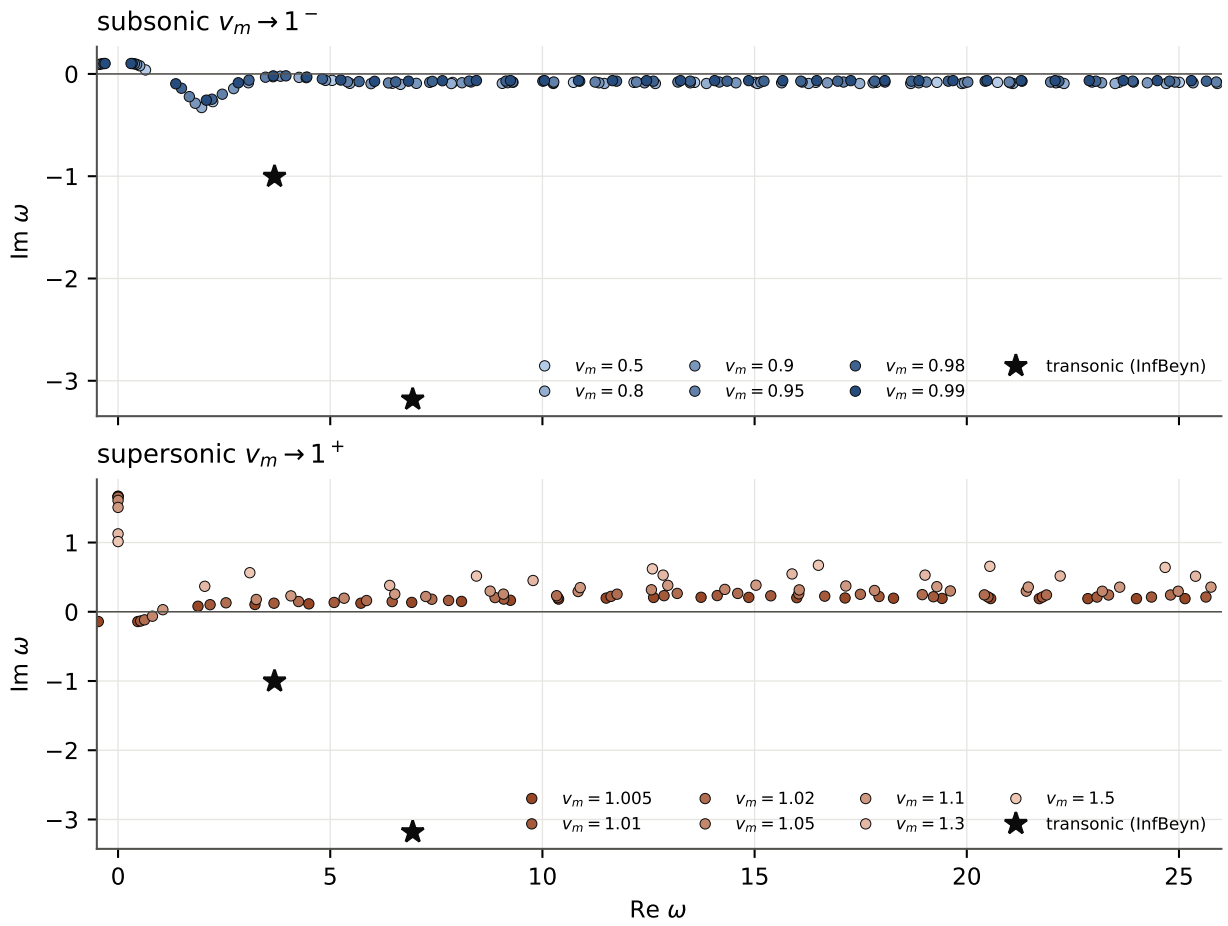


Figure 5: Certified InfBeyn spectra (fixed-open) at the swept v_m (darker closer to 1), with the transonic eigenvalues (stars).

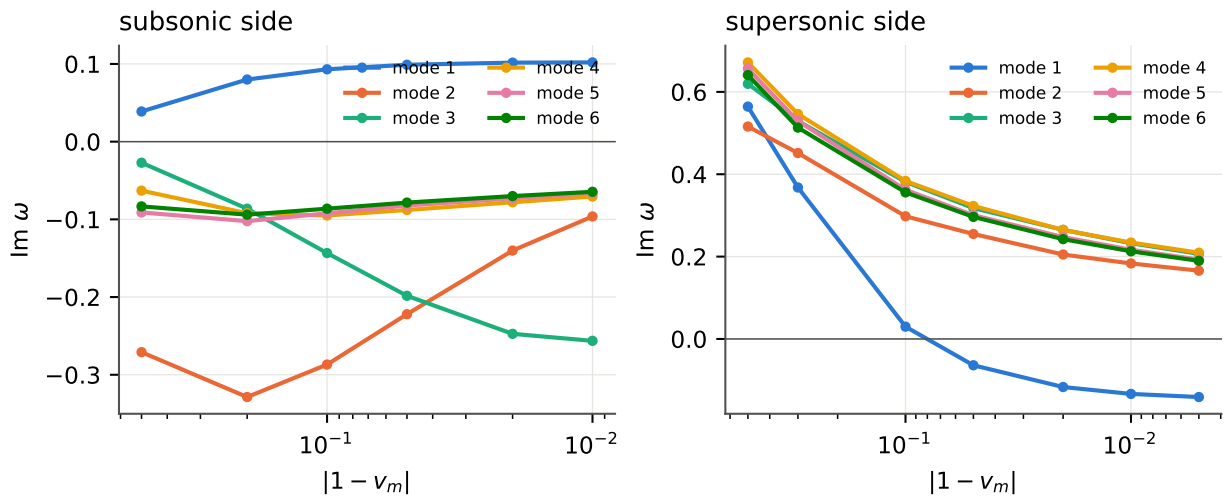


Figure 6: Growth rates of the six lowest modes with $\mathrm{Re}\,\omega > 0.05$ along the InfBeyn sweep, tracked by nearest neighbour. The sweep is much coarser than the recommended grid (steps of 0.2 between $v_m = 1.5$, 1.3, 1.1, against $\Delta v_m \approx 0.03$ needed there), so the mode labels may swap. The drop of supersonic mode 1 below zero is confirmed by the collocation continuation: it crosses between $v_m = 1.080$ and 1.085 (Sec. 4). The purely imaginary supersonic mode ($\mathrm{Re}\,\omega = 0$) is excluded here; its growth rate rises from 1.01 to 1.67.

A mirrored pattern near the sonic point (fixed-open). Since the spectrum is symmetric under $\omega \mapsto -\bar{\omega}$, only $\mathrm{Re}\,\omega > 0.05$ is considered here, and the purely imaginary supersonic mode is set aside. Modes are numbered by increasing $\mathrm{Re}\,\omega$. Close to $v_m = 1$ the two sides are mirror images of each other:

- Subsonic: the bulk of the spectrum is damped, mode 1 (the ground mode) is unstable ($\mathrm{Im}\,\omega_1 \approx +0.10$), and $\mathrm{Im}\,\omega_2 \to 0^-$.
- Supersonic: the bulk of the spectrum grows, mode 1 is stable ($\mathrm{Im}\,\omega_1 \approx -0.14$), and $\mathrm{Im}\,\omega_2 \to 0^+$.

Mode 2 looks as if it is about to cross the real axis, but down to $|1 - v_m| = 10^{-3}$ it has not crossed on either side. Its growth rate decreases like a power of $|1 - v_m|$ (Table 3); least-squares fits of $\ln|\mathrm{Im}\,\omega_2|$ against $\ln|1 - v_m|$ over $|1 - v_m| \le 0.02$ give

$$\mathrm{Im}\,\omega_2 \approx -1.12\,(1 - v_m)^{0.53} \quad (v_m \to 1^-), \qquad \mathrm{Im}\,\omega_2 \approx +0.56\,(v_m - 1)^{0.37} \quad (v_m \to 1^+). \tag{4}$$

The subsonic side is close to a $\sqrt{1 - v_m}$ law ($\mathrm{Im}\,\omega_2/\sqrt{1 - v_m}$ stays between -0.99 and -0.92). On the supersonic side the same ratio grows from 0.92 to 1.31, so the approach is slower than a square root. Extrapolated, both fits reach $\mathrm{Im}\,\omega_2 = 0$ exactly at $v_m = 1$: mode 2 becomes neutral in the limit, from below on the subsonic side and from above on the supersonic side. This is an extrapolation from $|1 - v_m| \ge 10^{-3}$ (the continuation at $v_m = 0.999$ did not converge for mode 2 and is left out). A log-log fit on the fine grid of Sec. 4, and a check of whether mode 3 behaves the same way closer to $v_m = 1$, would show whether $v_m = 1$ is a single neutral point or an accumulation of them.

v_m	$\mathrm{Im}\,\omega_1$	$\mathrm{Im}\,\omega_2$	$\mathrm{Im}\,\omega_2/\sqrt{1-v_m}$	v_m	$\mathrm{Im}\,\omega_1$	$\mathrm{Im}\,\omega_2$	$\mathrm{Im}\,\omega_2/\sqrt{v_m-1}$
0.980	0.102	−0.140	−0.99	1.020	−0.117	0.130	0.92
0.985	0.102	−0.120	−0.98	1.010	−0.133	0.102	1.02
0.990	0.102	−0.096	−0.96	1.0075	−0.137	0.092	1.07
0.9925	0.102	−0.082	−0.95	1.005	−0.141	0.080	1.13
0.995	0.101	−0.067	−0.94	1.003	−0.143	0.066	1.20
0.997	0.101	−0.051	−0.93	1.002	−0.144	0.056	1.26
0.998	0.100	−0.041	−0.92	1.001	−0.144	0.042	1.31

Table 3: Growth rates of the two lowest modes with $\mathrm{Re}\,\omega > 0.05$ near the sonic point, from the collocation continuation (throat-clustered grid, $N = 300$, reproduced at $N = 400$ to 10^{-6}). They agree with the certified InfBeyn sweep at $v_m = 0.99$, 1.005, 1.01, 1.02.

Away from $v_m = 1$ the mirror breaks. The subsonic ground mode is unstable for every v_m checked, down to $v_m = 0.05$ ($\mathrm{Im}\,\omega_1 = 0.0025$, roughly $0.05\,v_m$ for small v_m; plain-grid collocation, $N = 300$ against 400). The constant-profile limit has every mode damped, so the profile variation itself destabilizes the ground mode. The stable supersonic mode 1 instead exists only in a thin layer: $\mathrm{Im}\,\omega_1$ crosses zero between $v_m = 1.080$ (-0.008) and 1.085 ($+0.001$), and for larger v_m (checked up to 3) every mode with $\mathrm{Re}\,\omega > 0$ is unstable. The purely imaginary supersonic mode also has no subsonic counterpart: for constant v_0 the subsonic real parts are shifted by half a spacing, so no subsonic eigenvalue sits on the imaginary axis.

5 Summary

- **InfBeyn** is ready for $v_m \neq 1$ once the grid is clustered at the throat. That change is in `common.py` (`mapped_problem`); without it the method fails below $|1 - v_m| \approx 0.05$.
- **Frobenius**: the regularity condition is the right physics. Imposed as a collocation row inside InfBeyn it gives certified transonic eigenvalues. The thesis shooting results are correct only for the first mode, and modes 3–8 are not eigenvalues.
- **Grid**: logarithmic in $|1 - v_m|$, about 23 points per decade per side. That is 80 scans to reach $|1 - v_m| = 10^{-2}$ (about 30 minutes now) or 124 to reach 10^{-3} (one to two hours, after the solution-cache memory fix in `scan()`).
- **Open point**: the limits $v_m \to 1^{\pm}$ approach cusped profiles, and their spectra differ from the accelerating transonic spectrum. Deciding which object "the spectrum at the sonic point" should be is a physics question to settle before running the full sweep.

Files. `common.py` contains the profiles, the mapped problem, `TransonicInfBeyn` and collocation. `scan_sweep.py` runs the mapped InfBeyn sweep, `scan_timing.py` the plain-grid timing, `transonic_scan.py` the transonic problem, `frobenius_check.py` the series and re-shooting check, and `continuation.py` the collocation continuation. `make_figures.py` produces all figures and tables.

References

[1] H. Feng, *Spectral stability of plasma flows in the magnetic nozzle*, M.Sc. thesis, University of Saskatchewan, 2024.

[2] M. J. Colbrook and A. Townsend, *Avoiding discretization issues for nonlinear eigenvalue problems*, arXiv:2305.01691, 2023.